Hamit Efe Çinar

Research Interests

I study how algorithms and incentives can be designed and tuned automatically. My work spans algorithm configuration and selection, using machine learning and empirical performance models to make solvers for hard combinatorial problems faster, and algorithmic game theory, where computation meets strategic behaviour in mechanism design and multi-agent settings. I aim to combine formal reasoning with learning to build systems that are efficient, interpretable, and human-compatible.

Education

University of British Columbia — Vancouver, Canada
Ph.D. in Computer Science
Supervisor: Prof. Kevin Leyton-Brown — Sep 2026 - Present

- Research areas: Algorithm Configuration, Algorithmic Game Theory

Sabancı University — Istanbul, Turkey
B.S. in Computer Science and Engineering, B.S. in Industrial Engineering
Minor in Mathematics — Sep 2021 - June 2026

- GPA: 3.98/4.00, Top 3 GPA in the Faculty of Engineering and Natural Sciences
- Full Tuition Scholarship

Uppsala University (Erasmus Exchange) — Uppsala, Sweden
Computer Science and Mathematics — Jan 2024 - June 2024

Research Experience

CON-NET: Platform-Agnostic Bot Detection Model
TUBITAK & CHIST-ERA funded project
Supervisor: Dr. Onur Varol — Jan 2025 - Feb 2026

- Designing and implementing a platform-agnostic bot detection model
- Implementing account-based and digital-dna based VAE models
- Feature engineering for cross-platform compatibility and implementation of novel approaches (e.g. L2P and SpringRank) to avoid distribution shifts

Partial Observable Multi-Agent Path Finding with Constrained Communication
Erasmus Internship, University of Luxembourg @ ICR Group
Supervisors: Prof. Leon van der Torre & Dr. Pere Pardo — July - Sep 2025

- Designed a framework to solve a novel variant of the Multi-Agent Path Finding (MAPF) problem featuring partial observability, decentralization, and anonymous task-sequential requirements with constrained communication
- Modeled observation and communication using dynamic epistemic logic and employed Answer Set Programming for collaborative planning under uncertainty
- Developed a simulation script to compare performance across different exploration strategies and map configurations

Solving Matching Problems Using AI Methods
PURE Project, Sabancı University @ CogRobo Lab
Supervisor: Prof. Esra Erdem — Sep 2024 - Jan 2025

- Designed a student-to-department assignment system using Gale-Shapley algorithm and Constraint Programming to balance student preferences, diversity goals, and department capacities

Automatic Transcription of Ottoman Documents (AKIS)
TUBITAK funded project, Sabancı University @ DH Lab & VPA Lab
Supervisors: Prof. Berrin Yanıkoglu & Dr. Esma Bilgin Tasdemir — Aug 2023 - Jan 2024

- Enhanced image segmentation capabilities of existing R-CNN model and achieved significant improvement in model accuracy for document transcription

A Digital Analysis of an Early Ottoman Chronicle
PURE Project, Sabancı University @ DH Lab
Supervisors: Dr. Inanç Arın & Dr. Mehmet Kuru — July - Sep 2023

- Created geo-visualizations and interactive data analyses of Asıkpasazade's Chronicle
- Performed NLP analyses with BERT and BERTopic to explore sentiment and themes
- Developed an LLM-based chatbot using GPT-3.5 to enable dynamic user interaction with the chronicle

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Industry Experience

Legotize: 3D to LEGO Converter
Valensas Software, Supervisor: Akın Idil — Aug - Sep 2024

- Designed a program to convert 3D models into pixelated LEGO representations using PyTorch3D.

Vinventory: Inventory Management App
Valensas Software, Supervisor: Moray Baruh — July - Sep 2024

- Built an inventory management app using Go, React.js, and PostgreSQL with Azure AD authentication. Integrated CI/CD pipelines with GoReleaser and deployed on Kubernetes with Helm Charts.
- The app is currently being used as a tool inside the company.

OREDATA Data Science Intern | Istanbul, Turkey — Jan 2023 - Feb 2023

- Experimented with and compared different tumor-detection ML and DL models using scikit-learn. Practiced exploratory data analysis leveraging Kaggle datasets.

Teaching Experience

Teaching Assistant: Introduction to Artificial Intelligence (AI 100, University of British Columbia, Sep 2026 - present); Theory of Computation (Prof. Kemal Inan, Jan 2026 - June 2026); Algorithms (Prof. Esra Erdem, Jan 2025 - Jan 2026); Machine Learning (Dr. Onur Varol, Sep 2024 - Jan 2025); Data Structures (Dr. Gülsen Demiröz, Sep 2023 - Jan 2024).

Awards and Honors

- **Ranked 139th** among 2,4+ million students (**884th in STEM category**) in Turkey's national university entrance exam — 2021
- **Full Tuition Scholarship** and Monthly Stipend, Sabanci University — 2021
- **TUBITAK 1001 Undergraduate Scholarship**, Research Trainee on a TUBITAK-funded project with monthly stipend — Aug 2023 – Jan 2024
- **Top 3 GPA**, Faculty of Engineering and Natural Sciences, Sabanci University — 2026

Skills

Languages: Turkish (Native), English (IELTS: 8.0/9.0), German (Basic)

Programming: `Python`, `C/C++`, `Go`, `JavaScript`, `Java`, `Verilog`

ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NetworkX, Matplotlib, Seaborn, PyTorch3D

Tools & Infrastructure: Git, Docker, Kubernetes, Flask, MERN Stack, Unix/Linux CLI, LaTeX, Markdown

Databases: PostgreSQL, MySQL, NoSQL

Extra-curricular Activities

Google Developer Student Club: Led workshops on TensorFlow, Selenium and Git/GitHub for 2-3 hours each.

Literature Club: Board member of the Literature Club of Sabancı University for 4 years, the last year as the President. For the last 3 years, I was the coeditor of the club's semiannual literature fanzine, "harmoni.".

Social Responsibility: Volunteered in shoreline cleanups, Gender and Memory Walks, and botanic garden projects.